Akoizo Original Practice - 2027 Rotation Form 1

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Alpha-helix H bonds mainly involve…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only neighboring side-chain colors

B. Backbone groups

C. Only sulfur atoms

D. Only DNA bases

Answer: _____

2. (2 points) Attachment and fusion are…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only DNA processes

B. Absent in all enveloped viruses

C. Different entry-related steps

D. Always identical

Answer: _____

3. (2 points) Rotation changes…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Primary sequence

B. Chemical bonds automatically

C. PDB identifier

D. Viewpoint

Answer: _____

4. (2 points) Quaternary structure requires…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. No interfaces

B. Multiple polypeptide subunits

C. Exactly four atoms

D. Only one residue

Answer: _____

5. (2 points) An epitope is…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Every viral genome

B. A unit of salinity

C. A recognized molecular feature

D. The entire immune system

Answer: _____

6. (2 points) Color by structure shows…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Measured free energy

B. Every atom’s charge

C. A diagnosis

D. A structural classification

Answer: _____

7. (2 points) Denaturation always cleaves peptide bonds?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only at neutral pH

B. No

C. Always

D. Only for helices

Answer: _____

8. (2 points) Binding alone proves neutralization?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only for any antibody

B. Only in a diagram

C. No

D. Always

Answer: _____

9. (2 points) The built-in lab executes full Jmol?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

A. Yes

B. Only at maximum zoom

C. Only for HA

D. No

Answer: _____

10. (4 points) Compare motif and domain.

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

Answer: _____
_____
_____

11. (2 points) The assigned interval runs from residue 153 through 252, inclusive. How many numbered residue positions are included?

Answer: _____
_____
_____

12. (2 points) At the specified scale of 2 cm per residue, what backbone material length corresponds to 100 residues before folding?

Answer: _____
_____
_____

13. (2 points) State the assigned 2027 structure identifier, chain and residue interval, and distinguish these identifiers from a claim that all residues are experimentally resolved.

Answer: _____
_____
_____

14. (2 points) Explain the functional relevance of hemagglutinin binding to sialic-acid receptors in influenza entry.

Answer: _____
_____
_____

15. (2 points) Why does identifying an alpha helix require backbone geometry rather than simply finding a hydrophobic sequence stretch?

Answer: _____
_____
_____

16. (2 points) A synthetic structure worksheet places two oppositely charged side chains close together. Describe a possible stabilizing interaction and a qualification.

Answer: _____
_____
_____

17. (2 points) Explain why displaying a ligand-binding pocket without identifying chain and residue numbering can mislead a model reviewer.

Answer: _____
_____
_____

18. (2 points) What is the distinction between a displayed molecular contact and a proven functional interaction?

Answer: _____
_____
_____

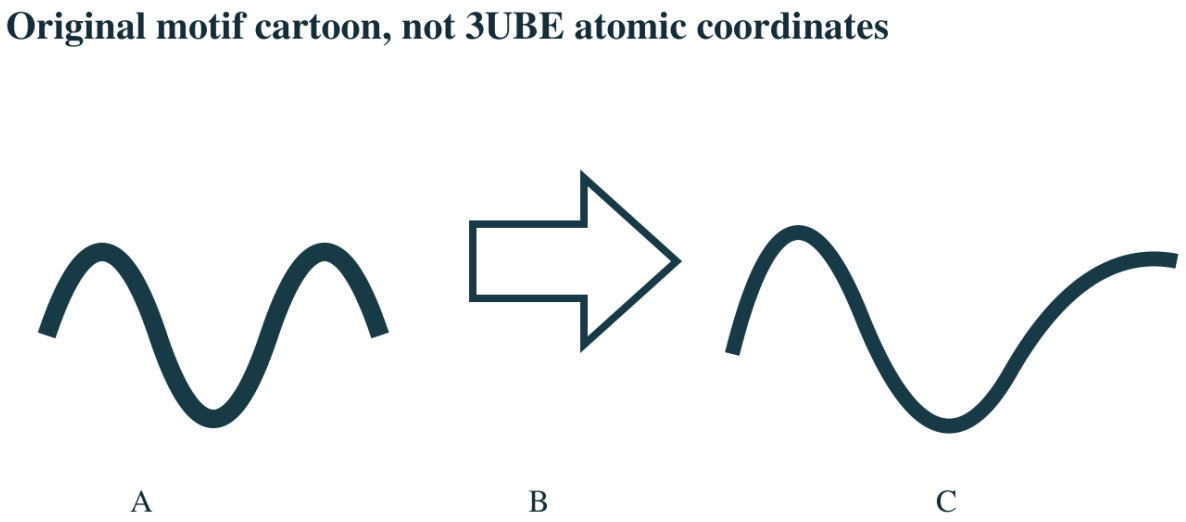
19. (2 points) A model includes a decorative feature unrelated to protein structure or function. Explain why it does not improve its scientific explanation.

Answer: _____
_____
_____

20. (2 points) Explain why rotating a JUDE molecular view does not alter the primary sequence or chemical connectivity.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Identify the secondary-structure motif convention represented by B.



F1. Backbone motifs and selected residue interval (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. How many residue positions are included in the selected subinterval?

Refer to F1. Backbone motifs and selected residue interval (original synthetic teaching figure) shown with the first question in this figure group.

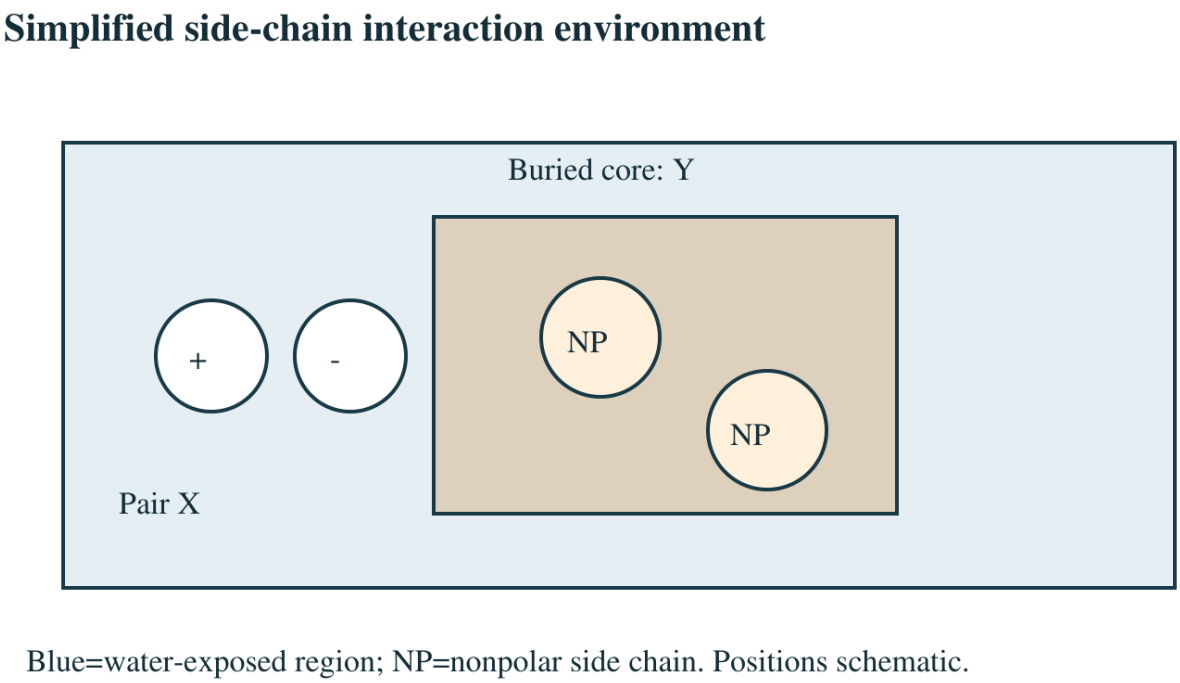
Answer: _____
_____
_____

23. (2 points) Use Figure F1. At2 cm per residue, what model backbone length represents that selected subinterval?

Refer to F1. Backbone motifs and selected residue interval (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. What stabilizing interaction could occur at pair X? State one condition that affects it.



F2. Charge and burial in a folded protein (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Explain why nonpolar side chains in Y can be favorable in an aqueous environment.

Refer to F2. Charge and burial in a folded protein (original synthetic teaching figure) shown with the first question in this figure group.

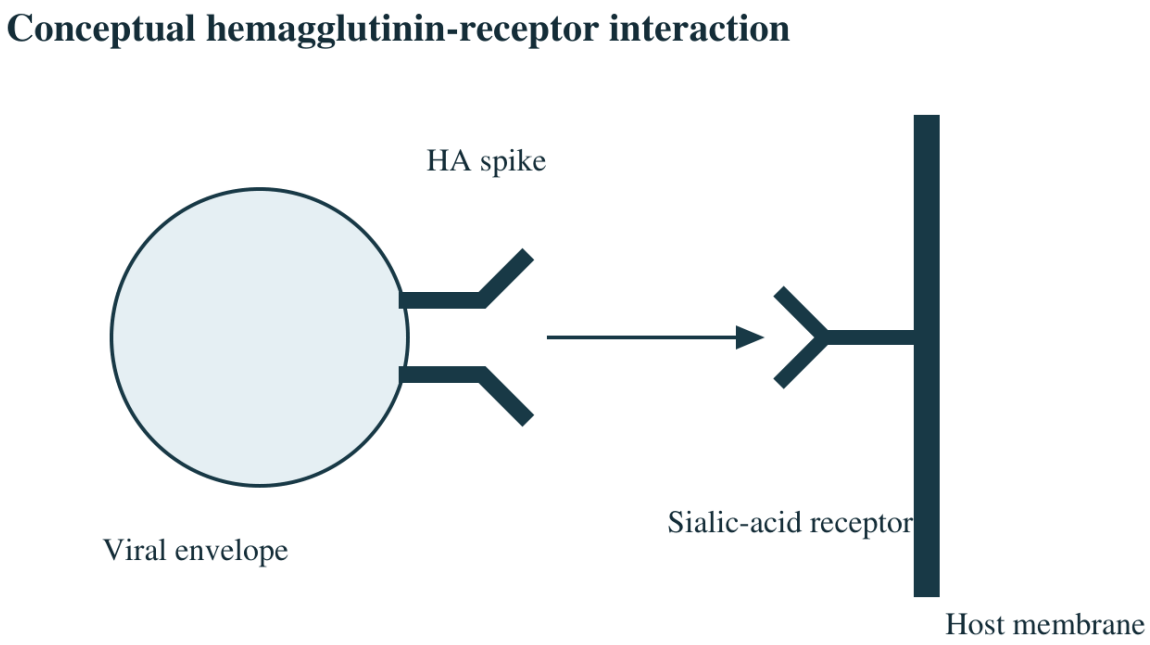
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Why could changing pH alter the displayed interaction at X?

Refer to F2. Charge and burial in a folded protein (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. [State/National] Which labeled viral feature directly participates in the depicted receptor attachment?



F3. Influenza attachment mechanism (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. [State/National] Predict how blocking the displayed receptor-binding interaction could affect this entry step.

Refer to F3. Influenza attachment mechanism (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

29. (2 points) Use Figure F3. [State/National] Why does this attachment drawing not prove that a bound particle has completed membrane fusion?

Refer to F3. Influenza attachment mechanism (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which recorded field distinguishes the intended chain from other chains in the same structure?

Illustrative JUDE selection record; not a captured application screen

Field	Recorded selection
Structure	3UBE
Chain	A
Residue range	153-252 inclusive
Display	Backbone cartoon
View action	Rotate camera
Coordinate check	Check missing residues before building

F4. Molecular-view selection record (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Does the recorded camera rotation change primary sequence or bonds? Explain.

Refer to F4. Molecular-view selection record (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Why is the final coordinate check important even when the numbered interval is correctly recorded?

Refer to F4. Molecular-view selection record (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____